

Abelti Beshana Belay

Senior Software Engineer | .NET · TypeScript / Next.js · Distributed Systems

+251 986 990 019 | abeltibbeshana@gmail.com | Addis Ababa, Ethiopia (UTC+3) | Open to Remote
github.com/ObsidianBeing | linkedin.com/in/abeltibbeshana | abeltib.vercel.app

PROFESSIONAL SUMMARY

Senior software engineer with 6+ years building and operating production systems across transport, fintech, payments, and education technology. Deep in ASP.NET Core (C#) and TypeScript/Next.js, with end-to-end ownership from database design and API architecture through deployment and production support. Currently maintain and extend platforms serving 17,000+ passengers per day. Held COO and CTO titles at two early-stage startups (under 12 people each) where the roles were hands-on rather than executive — I wrote and shipped the code, and deliberately returned to a full-time individual-contributor engineering track to keep building. Fully remote for the past year across distributed teams; comfortable with async communication, written specs, and self-directed delivery.

TECHNICAL SKILLS

Languages: C#, TypeScript, JavaScript, Python, Java, SQL, HTML5, CSS3

Backend: ASP.NET Core, .NET 8/9/10, Node.js, Express.js, Hono.js, NestJS, FastAPI, Next.js API Routes, Microservices, REST, GraphQL, gRPC-style service communication

Frontend: React.js, Next.js 14/15/16, React Native, Blazor / WebAssembly, Tailwind CSS, Redux, Three.js

Databases: PostgreSQL, Microsoft SQL Server, MySQL, MongoDB, Neo4j, Redis, Prisma ORM, Entity Framework Core, schema design, query optimization, data migration

Cloud & DevOps: Docker, Git, GitHub, GitHub Actions, CI/CD pipelines, Vercel, Supabase, Linux server administration, Nginx, application monitoring and performance tuning

AI & Optimization: LLM/API integration (OpenAI, Anthropic, Hugging Face), Google OR-Tools CP-SAT constraint solving, retrieval-augmented search, data pipelines, model fine-tuning

Payments & Integrations: M-Pesa, Stripe, Telebirr, SantimPay, third-party API integration, webhook/callback handling, OTP verification, idempotent transaction design, JWT and OAuth authentication

Architecture & Practice: Clean Architecture, Domain-Driven Design, RBAC/ABAC authorization, multi-tenancy, Agile/Scrum, code review, technical documentation, Swagger/OpenAPI, Postman

PROFESSIONAL EXPERIENCE

Senior Software Engineer (Full-time, Remote)

Mar 2025 – Present

BS Technologies PLC — intercity transport technology operator

Remote — Addis Ababa, Ethiopia

- Own backend and full-stack development for a web and mobile transport platform managing drivers, vehicles, routes, and fares for 17,000+ passengers per day across a national terminal network.
- Architected and shipped an internal scheduling and operations platform (Next.js 15 + .NET APIs + PostgreSQL) covering ticketing, biometric attendance, employee management, station oversight, and route administration, replacing manual spreadsheet workflows.
- Designed a constraint-based driver and vehicle scheduling engine using Google OR-Tools CP-SAT, cutting schedule preparation from a multi-day manual process to under an hour while enforcing fairness, rest-period, and capacity constraints.
- Led migration of legacy services to the current .NET framework, resolving performance bottlenecks and measurably improving API response times and system stability under peak booking load.
- Rebuilt the online booking administration dashboard in Next.js with AI-assisted sorting and filtering, reducing the time operations staff spend locating and reconciling bookings.
- Built an HR and payroll web system (Next.js, Prisma, PostgreSQL, ABAC, phone OTP auth) covering ~47 employees across branch locations, replacing an error-prone Excel registry.

- Automated ingestion of regulator trip-reporting data via a Python pipeline against a JSON API, collecting 10,000+ trip records and generating per-terminal reconciliation reports.
- Administer application servers, monitor system health, and tune performance for high availability across web and mobile platforms.

Instructor, Software Engineering & Computer Science (Part-time)

HiLCoE School of Computer Science and Technology

Jun 2024 – Present
Addis Ababa, Ethiopia

- Teach undergraduate and graduate software engineering and computer science courses, developing lecture materials, lab exercises, and project-based assessments.
- Contribute to curriculum development, aligning course content with current industry practice in web architecture, databases, and software design.
- Mentor students through full software development lifecycles, from requirements analysis to deployment.

Operations Manager & Lead Instructor

Malefia Training Institute — early-stage training startup

Feb 2024 – Mar 2025
Addis Ababa, Ethiopia

- Designed and delivered a full-stack web development curriculum covering the MERN stack, mobile development, Python, Java, and .NET/C#, including slides, recorded lectures, and documented reference repositories.
- Built and customized the institute's WordPress-based Learning Management System, improving course delivery and student engagement.
- Directed daily operations, staffing, and workflow design as the organization expanded from short courses into corporate training services.

Instructor & Developer — Full-Stack Web Development

Ethio Bytes IT and Digital Solutions

Jan 2023 – Sep 2023
Addis Ababa, Ethiopia

- Delivered a MERN-stack (MongoDB, Express.js, React.js, Node.js) full-stack curriculum with hands-on projects and fully documented source code.
- Contributed to the center's Learning Management System using HTML, CSS, JavaScript, and PHP.

Chief Operating Officer (hands-on engineering role)

Prestige Software Solution Technologies PLC — software startup, under 12 employees

Nov 2020 – Dec 2022
Addis Ababa, Ethiopia

- Built and deployed in-house and client systems, including a Point-of-Sale platform and a disability support platform — personally responsible for architecture, backend APIs, and database design.
- Prototyped a sign-language-to-Amharic translation system combining computer vision and NLP techniques.
- Ran delivery for concurrent client projects: scoping, staffing, progress tracking, and quality assurance, in a team small enough that every leadership responsibility came with direct implementation work.

Chief Technology Officer → IT Project Manager (concurrent engagement)

My Car Trading PLC — early-stage automotive startup, under 10 employees

Feb 2021 – Nov 2022
Addis Ababa, Ethiopia

- Led the full product lifecycle of a web and mobile roadside-assistance platform (ASP.NET Core, MSSQL, Blazor, Clean Architecture) from concept through deployment and post-launch optimization.
- Architected the system end to end — database design, backend APIs, and frontend/mobile implementation — writing production code throughout rather than managing from above.
- Analyzed the legacy platform, identified performance bottlenecks, and rebuilt on a modern, scalable stack.

SELECTED PROJECTS

M-Pesa Mobile Wallet Payment Gateway / *ASP.NET Core Web API, C#, PostgreSQL, JWT*

- Built a secure payment gateway integrating M-Pesa mobile wallet transactions with internal systems: transaction initiation, callback handling, OTP verification, and real-time status reconciliation with idempotent retry handling for failed callbacks.

Habexa — Investment Education & Paper-Trading Platform / *Next.js, TypeScript, PostgreSQL, Prisma*

- Designing a fintech platform for the Ethiopian Securities Exchange delivering paper trading and investment education to retail and diaspora investors, including market data modeling, portfolio simulation, and multi-repository system architecture.

University Timetable Scheduling System / *Next.js, Prisma, TypeScript, Python FastAPI, Google OR-Tools CP-SAT*

- Built a production scheduling system that solves multi-constraint academic timetabling via a dedicated CP-SAT solver microservice, with automated clash detection across student cohorts, rooms, and instructors.

Driving License Training Center Platform / *.NET Microservices, MSSQL, Docker, Ocelot API Gateway*

- Designed a microservices-based management system handling student enrollment, lesson scheduling, payments, and exam tracking, with service isolation and gateway routing for modular expansion.

Church Management System / *Next.js 15, MongoDB, REST APIs*

- Built a church administration platform for a US-based congregation covering member management, event scheduling, contribution tracking, and attendance logging.

Event Registration & Delivery Platform (Run to Hold 5K) / *React Native, Next.js, MongoDB*

- Delivered a hybrid mobile and web solution for event registration, merchandise ordering, and delivery management with real-time status updates.

EDUCATION

Bachelor of Science in Computer Science

HiLCoE School of Computer Science and Technology

Sep 2016 – Sep 2021

Addis Ababa, Ethiopia

CERTIFICATIONS & ADDITIONAL TRAINING

- Excellency Training — Dereja Academy, Addis Ababa (2021): software development, project management, analytical thinking, and professional communication.

LANGUAGES

Amharic — Native | English — Full professional proficiency (written and spoken)